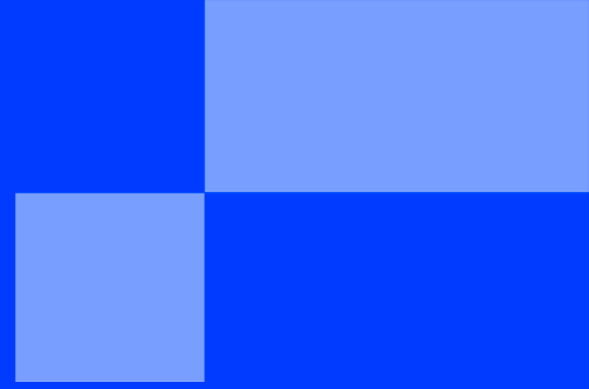
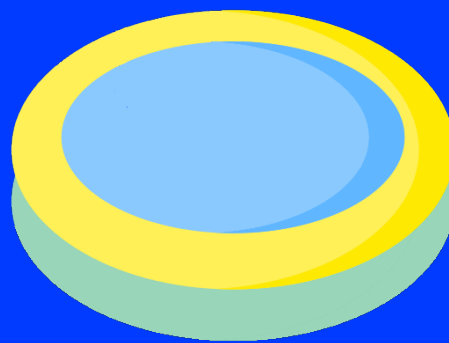




# PREPARATION BOOTCAMP

< FUNCTIONS />



# PREPARATION BOOTCAMP

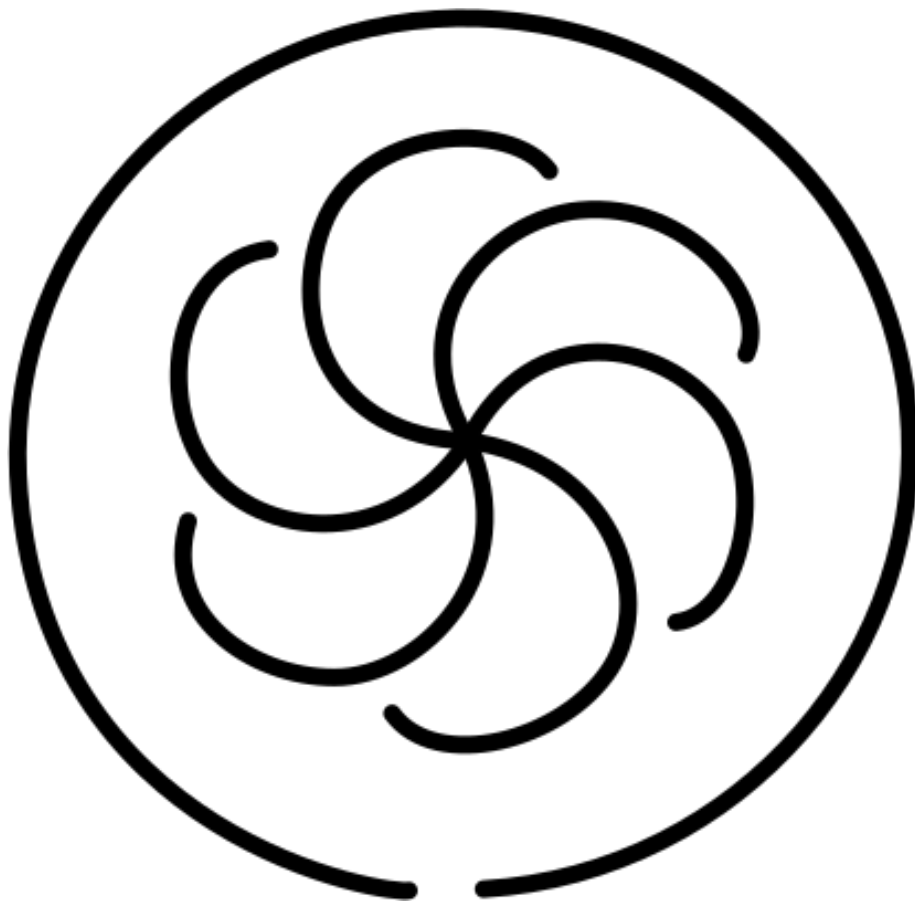
## GAME



### Code wars

In addition to the tasks below, you must go as far as possible in [this code wars collection](#). Try to solve them from the first one to the last one, without skipping challenges!

Work on it as soon as you have a bit of time, or whenever you need a break in your day!



## User-defined functions

### Task 1.1



Dig this piece of code and try to predict the values of the output. Then, run it to check it.

```
def f1():  
    return 42  
def f2(x):  
    return 2 * x  
print(f1(), f2(5) + f1())
```

### Task 1.2



Using the following functions, display a lettuce-tomato-double ham sandwich in your terminal.

```
def bread():  
    print("</////////>")  
def lettuce():  
    print("~~~~~")  
def tomato():  
    print("0 0 0 0 0")  
def ham():  
    print("=====")
```

### Task 1.3



Write a function that takes a number of sandwiches to prepare as a parameter. Then displays as many sandwiches as requested if the parameter is correct, and `I can't do this!` if it's not (such as `3.14`).

### Task 1.4



Add a parameter to provide the possibility for a veg sandwich (double vegetables + no ham). If this option isn't specified, the sandwich must be a lettuce-tomato-double ham one by default.

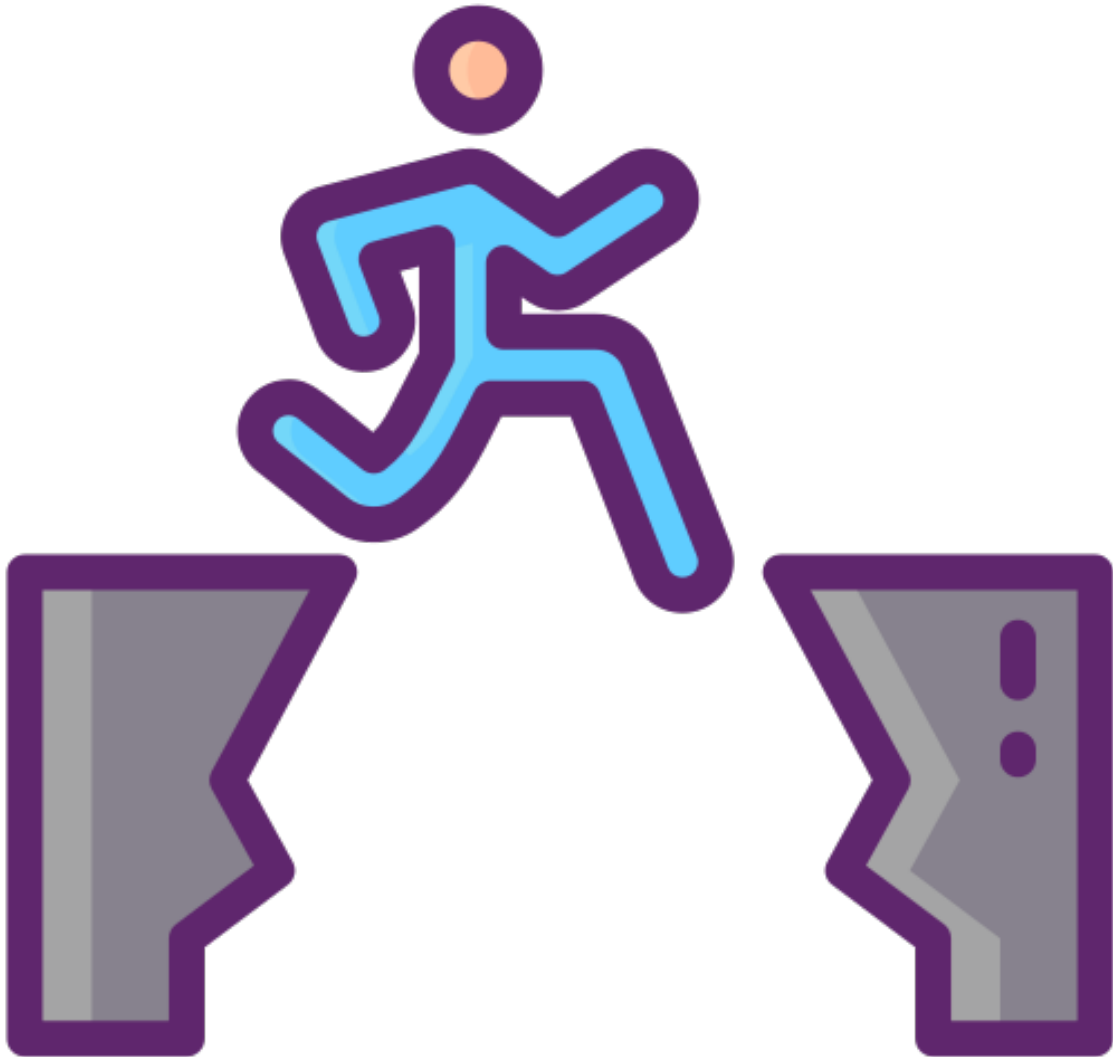


## CHALLENGE

Write a little program that computes the power function as fast as possible.

How long does your program take to compute  $42^{84}$ ? and  $42^{168}$ ?

Compare your times and your code with your neighbors.



You are not allowed to use the system built-in functions yet!

## Recursion

### Task 2.1



Write a **recursive** function that computes the sum of all integers from 1 to  $n$ , a given parameter.



The parameter 42 should return 903.

### Task 2.2



Write a **recursive** function that prompts the user for a string of characters, strips out the spaces and punctuation signs, lowercases the string, then tests if it is a palindrome.



It should return "True" for never odd or even and A Santa Lived As a Devil at NASA.

### Task 2.3



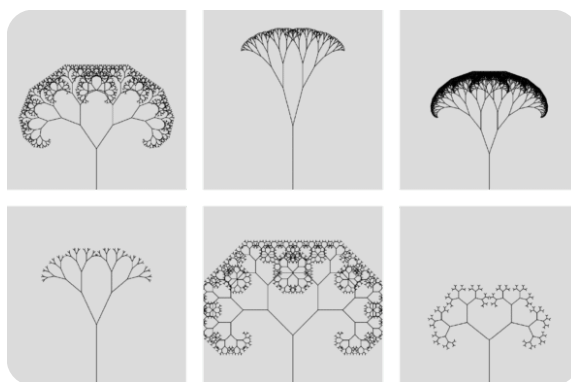
Write a **recursive** program that lists all the files and directories in the current directory, as well as all files and directories in its sub-directories and so on.



It should behave similarly to the `ls -R` command.



Check out the Python **operating system** interface.



## Built-in functions

Imagine you're building a house, and you decide to carve your own bricks from scratch.



Sounds like a lot of work, right? Why reinvent the wheel? Why spend hours shaping custom functions when something already got your back!?

Python's built-in functions will make your life easier, like a set of powerful tools.

### Task 3.1



Use some built-in functions to get:

- ✓ the absolute value of 2, -2 and -3e4;
- ✓ the rounded value of 3.14159 to two decimal places;
- ✓ the maximum value of [-42, 3e2, 666].

### Task 3.2



Apply the `min()` built-in function on the string `Beautiful is better than ugly..`  
What do you observe? What do you make of it?

### Task 3.3

---



Using a built-in function, compute the value of 73 to the power 73.

### Task 3.4

---



What are the built-in functions that return:

- ✓ `True` if there is at least one true element in a list?
- ✓ `True` if there is zero false element in a list?

### Task 3.5

---



Using a built-in function, calculate the sum of `L1 + L2 + L3 + L4` where:

- ✓ `L1 = [1, 2, 3, 4];`
- ✓ `L2 = [5, 7, 9, 32];`
- ✓ `L3 = [23, 13, 17, 14, 16, 309];`
- ✓ `L4 = [10, 20, 30, 40].`

### Task 3.6

---



Find an easy way to sort the names in this list `["Joe", "William", "Jack", "Averell"]`:

- ✓ from shortest to longest;
- ✓ from longest to shortest.



v 2.2

{EPITECH}